

VECTOR SPACE

A vector space over $\mathbb{R}$ is a set V with 2 operations

$+: V \times V \rightarrow V, (x, y) \mapsto x + y$

“ADDITION”

$\cdot: \mathbb{R} \times V \rightarrow V, (\alpha, x) \mapsto \alpha \cdot x$

“MULTIPLICATION WITH A SCALAR”

that satisfy the following properties:–

1. $(V, +)$ is a commutative group, i.e.,

i) $+$ is associative

ii) Every element of V has an inverse element w.r.t $+$

iii) It has an identity element

iv) $+$ is commutative

2. $1 \cdot x = x \quad \forall x \in V$

3. $\alpha \cdot (x + y) = \alpha \cdot x + \alpha \cdot y \quad \forall \alpha \in \mathbb{R}, x, y \in V$

4. $(\alpha_1 + \alpha_2) \cdot x = \alpha_1 \cdot x + \alpha_2 \cdot x \quad \forall \alpha_1, \alpha_2 \in \mathbb{R}, x \in V$

5. $(\alpha_1 \alpha_2) \cdot x = \alpha_1 \cdot (\alpha_2 \cdot x) \quad \forall \alpha_1, \alpha_2 \in \mathbb{R}, x \in V$

SPAN OF VECTORS

The span of the vectors $v_1, v_2, \dots, v_n$ is the set of all linear combinations of $v_1, v_2, \dots, v_n$

$$\text{span}(v_1, v_2, \dots, v_n)$$

Q: Show that $\text{span} \left(\begin{bmatrix} 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) = \mathbb{R}^2$

$\Rightarrow$ Let $c_1, c_2 \in \mathbb{R}$

$$c_1 \begin{bmatrix} 2 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2c_1 \\ c_2 \end{bmatrix} \in \mathbb{R}^2$$

$\Rightarrow$ Conversely let $w \in \mathbb{R}^2$

$$w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \frac{w_1}{2} \begin{bmatrix} 2 \\ 0 \end{bmatrix} + w_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \uparrow \mathbb{R} \quad \quad \uparrow \mathbb{R}$$

LINEAR INDEPENDENCE

A list of vectors is linearly independent if none of the vectors can be written as a linear combination of others.

Proposition:

$v_1, v_2, \dots, v_n$ are L.I. $\iff \forall \alpha_1, \alpha_2, \dots, \alpha_n \in \mathbb{R}$

s.t. $\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n = 0$, then $\alpha_1 = \alpha_2 = \dots = \alpha_n = 0$.

LINEAR TRANSFORMATION

Given 2 vector spaces V & W , $L: V \rightarrow W$ is a linear transformation if

$\cdot L(v_1 + v_2) = L(v_1) + L(v_2) \quad \forall v_1, v_2 \in V$

$\cdot L(\alpha v) = \alpha L(v) \quad \forall v \in V$

Ex: $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2, \begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} x \\ 5 \end{bmatrix}$

$$L \left(\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \right) = L \left(\begin{bmatrix} x_1 + x_2 \\ y_1 + y_2 \end{bmatrix} \right) = \begin{bmatrix} x_1 + x_2 \\ 5 \end{bmatrix}$$

$$L \left(\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} \right) + L \left(\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} \right) = \begin{bmatrix} x_1 \\ 5 \end{bmatrix} + \begin{bmatrix} x_2 \\ 5 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ 10 \end{bmatrix}$$

$$L \left(\alpha \begin{bmatrix} x \\ y \end{bmatrix} \right) = L \left(\begin{bmatrix} \alpha x \\ \alpha y \end{bmatrix} \right) = \begin{bmatrix} \alpha x \\ 5 \end{bmatrix}$$

$$\alpha L \left(\begin{bmatrix} x \\ y \end{bmatrix} \right) = \alpha \begin{bmatrix} x \\ 5 \end{bmatrix} = \begin{bmatrix} \alpha x \\ 5\alpha \end{bmatrix} \Rightarrow \text{not a linear transform}$$

LINEAR TRANSFORMATIONS

A function from 1 vector space to another respecting the underlying (linear) structure of each space

- Let $\hat{i} : (1, 0)$ & $\hat{j} : (0, 1)$ be the basis vectors in the original vector space
- Now say a Linear Transform converts the 2 basis vectors as:

$$\hat{i} : (1, 0) \rightarrow (a, c) \quad \& \quad \hat{j} : (0, 1) \rightarrow (b, d)$$

- Now the linear transformation of any point in the original space can be expressed in terms of the linear transformation of the basis vectors

$$\text{Eg: } \vec{p} = q\hat{i} + r\hat{j}, \quad \hat{i} \mapsto \begin{bmatrix} a \\ c \end{bmatrix}, \quad \hat{j} \mapsto \begin{bmatrix} b \\ d \end{bmatrix}$$

$$\Rightarrow T(\vec{p}) = qT(\hat{i}) + rT(\hat{j}) = q \begin{bmatrix} a \\ c \end{bmatrix} + r \begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} q \\ r \end{bmatrix}$$

→ Linear Transform M^x

MATRICES AS LINEAR TRANSFORMATIONS & DETERMINANTS AS AREA OF TRANSFORMATION

- Any $n \times n$ matrix can be thought of as a transformation of the basis in 1 vector field to another
- Determinant gives a measure by which the area formed by the basis vectors scales on application of the transformation

Examples with diagrams:

$$\det \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} = 5, \quad \det \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} = -3, \quad \det \begin{bmatrix} 1 & 4 \\ -3 & 3 \end{bmatrix} = 15$$

SINGULAR & NON-SINGULAR SYSTEMS

⇒ **SINGULAR SYSTEM:** Redundant or contradictory

⇒ **NON-SINGULAR SYSTEM:** Complete

RANK

A measure of redundancy for the system: Maximal number of linearly independent columns/rows

⇒ Determinant of a Matrix with linearly dependent rows = 0

MATRIX ROW REDUCTION (GAUSSIAN ELIMINATION)

1. Write equations in the form of a Matrix
2. Use row manipulations to reach Row Echelon form
3. Row Echelon form → Row-Reduced Echelon form

ROW-ECHELON FORM

1. Rows consisting of only 0s are at the bottom
2. Leading entry of every non-0 row is to the right of that of every row above

**Rank of Matrix in Row Echelon Form: # of all non-0 rows or # of pivots*

ROW-REDUCED ECHELON FORM

1. It is in Row-Echelon form
2. The leading entry in each non-0 row is 1
3. Each column with a leading 1 has a 0 as all its other entries
4. Any number above a pivot is 0

RANK OF A MATRIX (Application $\rightarrow$ Image Compression (SVD))

The amount of information a system carries

$\Rightarrow$ For a matrix, the dimension of the solution space is related to its rank as:

$$\boxed{R = n - d} \longrightarrow \dim \text{ of } M^x$$

Examples:

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, d = 0, r = 2 \quad (\text{point}) \quad \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}, d = 1, r = 1 \quad (\text{line}) \quad \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, d = 2, r = 2 \quad (\text{plane})$$

RANK

The rank of a linear transform $L : V \rightarrow W$ is the dimension of the **IMAGE** of L

The set

$$\text{Im } L = L(V) = \{ w \in W \mid \exists v \in V \text{ s.t. } L(v) = w \}$$

$$\text{rank}(L) = \dim(\text{Im } L)$$

$$\text{Ex: rank of } L : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} x \\ 0 \end{bmatrix}$$

$$\text{Im } L = \left\{ \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \mid \exists \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}, L \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \right\}$$

$$L \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} v_1 \\ 0 \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \Rightarrow w_2 = 0$$

$$\Rightarrow \text{Im } L = \left\{ \begin{bmatrix} w_1 \\ 0 \end{bmatrix} \mid w_1 \in \mathbb{R} \right\}$$

$$\Rightarrow \text{rank } L = \dim(\text{Im } L) = 1$$

KERNEL OF A LINEAR TRANSFORMATION

The kernel (or nullspace) of a L.T. $L : V \rightarrow W$ is the set of vectors of V that are mapped to 0_W

$$\text{Ker}(L) = \{ v \in V \mid L(v) = 0_W \}$$

$$\text{Ex: Ker of } L : \mathbb{R}^2 \rightarrow \mathbb{R}^2, \begin{bmatrix} x \\ y \end{bmatrix} \mapsto \begin{bmatrix} x \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} \text{ s.t. } L \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow x = 0$$

$$\text{Ker}(L) = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2 \mid L \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\} = \left\{ \begin{bmatrix} 0 \\ b \end{bmatrix} \mid b \in \mathbb{R} \right\}$$

NOTE: $\dim(\text{Ker}(L)) = 1$

$$\dim(\text{Ker}(L)) + \text{rank } L = 1 + 1 = 2 = \dim(\mathbb{R}^2)$$

RANK-NULLITY THEOREM

Let V, W be 2 vector spaces & V be finite dimensional.

Let $L : V \rightarrow W$ be a linear transformation, then

$$\dim(\text{Ker}(L)) + \text{rank } L = \dim V$$

SUBSPACES

Let V be a vector space over $\mathbb{R}$. A non-empty subset $U \subset V$ is a subspace of V if U is a vector space over $\mathbb{R}$ w.r.t the same operations.

NOTE: A non-empty subset $U \subset V$ is a subspace of $V \iff \forall u_1, u_2 \in U, \alpha \in \mathbb{R}$ we have:

1) $u_1 + u_2 \in U$, 2) $\alpha u_1 \in U$

Examples of subspaces:-

- i) $\{0_V\}$ trivial subspaces of V
- ii) V
- iii) span of vectors of V
- iv) Kernel $\forall L : V \rightarrow W$ linear transforms
- v) Image

Theorem: Let V be a vector space of finite dimension & $U \subset V$ be a subspace of V , then

$$\dim U \leq \dim V$$

and

$$\dim U = \dim V \iff U = V$$

SURJECTIVE, INJECTIVE & BIJECTIVE FUNCTIONS

Let $f : E \rightarrow F$ be a function.

Surjective Fn: If all elements in F have a pre-image in V .

$$\rightarrow \forall y \in F, \exists x \in E \text{ s.t. } f(x) = y \quad i.e. \quad \boxed{\text{Im } f = F}$$

Injective Fn: If 2 diff. elements of E are mapped to 2 diff. elements of F .

$$\rightarrow \forall x_1, x_2 \in E \text{ s.t. } f(x_1) = f(x_2) \Rightarrow x_1 = x_2$$

Bijective Fn: Surjective + Injective

$\rightarrow$ If f is bijective, then $\forall y \in F \exists$ a unique $x \in E$ s.t. $f(x) = y$

$\rightarrow$ So if f is bijective, we can define the inverse function:

$$f^{-1} : F \rightarrow E, \quad f^{-1}(y) = x$$

Proposition: Let V, W be 2 vector spaces & $L : V \rightarrow W$ be a L.T.

- 1) $L(0_V) = 0_W$ (so $0_V \in \text{Ker}(L)$)
- 2) L is injective $\iff \text{Ker}(L) = \{0_V\}$

MATRICES AND LINEAR TRANSFORMATIONS

- Let A be an $m \times n$ M^x , then $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$, $x \mapsto Ax$ is a Linear Transform
- Let $L : E \rightarrow V$ be a linear transform where E, V are vector spaces of dimensions n & m .

$\rightarrow$ Let $\{a_1, a_2, \dots, a_n\}$ be a basis of E
 $\{b_1, b_2, \dots, b_m\}$ be a basis of V

$\rightarrow$ Apply L to each a_i , then we obtain a vector of V which can be written as a linear combination of $b_1, b_2, \dots, b_m$:

$$L(a_1) = \alpha_{11}b_1 + \alpha_{12}b_2 + \dots + \alpha_{1m}b_m$$

$$L(a_2) = \alpha_{21}b_1 + \alpha_{22}b_2 + \dots + \alpha_{2m}b_m$$

$$\vdots$$

$$L(a_n) = \alpha_{n1}b_1 + \alpha_{n2}b_2 + \dots + \alpha_{nm}b_m$$

$\rightarrow$ So we can define the M^x as:

$$B = \begin{bmatrix} \alpha_{11} & \alpha_{12} & \dots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \dots & \alpha_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{m1} & \alpha_{m2} & \dots & \alpha_{mn} \end{bmatrix}$$

Representation of L in the bases $\{a_1, \dots, a_n\}$ and $\{b_1, \dots, b_m\}$.

Example: Matrix Associated with Linear Transform

Ex: Find the M^x associated with the linear transform

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3, \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mapsto \begin{bmatrix} z \\ x \\ y \end{bmatrix}$$

in the standard basis of $\mathbb{R}^3$.

$$\begin{aligned} e_1 &= \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\ T(e_1) &= \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad T(e_2) = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad T(e_3) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \\ \Rightarrow B &= \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \end{aligned}$$

Interpretation for General Bases

Let $L : E \rightarrow V$ be a L.T. & B its representation in the bases $\{a_1, \dots, a_n\}$ of E & $\{b_1, \dots, b_m\}$ of V .

If $x \in E$ has coordinates $\lambda_1, \lambda_2, \dots, \lambda_n$ w.r.t $\{a_1, \dots, a_n\}$ then the coordinates of $L(x)$ are given by the M^x -vector multiplication:

$$B \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_n \end{bmatrix}$$

RANK OF A MATRIX ($m \times n$)

The rank of the associated Linear Transformation

$$A : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad x \mapsto Ax$$

$$\text{i.e., rank } A = \dim(\text{Im } A) \quad (\text{Range of } M^x A)$$

Remark: Since $Ax = x_1 A_1 + x_2 A_2 + \dots + x_n A_n$ (linear combination of the columns of A)

$$\Rightarrow \text{rank } A = \dim(\text{span}(A_1, A_2, \dots, A_n))$$

KERNEL OF A MATRIX

Defined as $\text{Ker } A \Rightarrow$ L.T., i.e.

$$\text{Ker } A = \{ x \in \mathbb{R}^n \mid Ax = 0 \} \quad (0 \text{ vector of } \mathbb{R}^m)$$

Ex: $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, calculate rank & kernel.

$$\rightarrow \text{rank } A = \dim \left(\text{span} \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \end{bmatrix} \right) \right) = 2$$

$$\text{So } A : \mathbb{R}^3 \rightarrow \mathbb{R}^2, \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mapsto A \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\begin{aligned} \text{Ker } A &= \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mid A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\} = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \mid \begin{bmatrix} y \\ 2z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\} \Rightarrow y = z = 0 \\ &\Rightarrow \text{Ker } A = \left\{ \begin{bmatrix} x \\ 0 \\ 0 \end{bmatrix} \mid x \in \mathbb{R} \right\} \end{aligned}$$

Remark: The rank-nullity theorem is verified:

$$\dim(\text{Ker } A) + \text{rank } A = 1 + 2 = 3$$

THEOREM

Let A be a square $n \times n$ M^x . Then the following are equivalent:

- i) $A : x \mapsto Ax$ is bijective
- ii) $A : x \mapsto Ax$ is surjective, i.e. $\text{range}(A) = \mathbb{R}^n$
- iii) $A : x \mapsto Ax$ is injective, i.e. $\text{Ker}(A) = \{0\}$

Proof: Since bijective means surjective & injective, it is enough to prove that i) $\Leftrightarrow$ ii) $\Leftrightarrow$ iii).

$\rightarrow$ First we prove ii) $\Rightarrow$ iii):

If $\text{range}(A) = \mathbb{R}^n$, then $\text{rank}(A) = n$. Then by rank-nullity theorem:

$$\dim(\text{Ker } A) + \underbrace{\text{rank } A}_{=n} = n \Rightarrow \dim(\text{Ker } A) = 0 \Rightarrow \text{Ker } A = \{0\}$$

$\rightarrow$ Next we prove iii) $\Rightarrow$ ii):

If $\text{Ker}(A) = \{0\}$, then $\dim(\text{Ker } A) = 0$. Then by rank-nullity theorem:

$$0 + \text{rank } A = n \Rightarrow \text{rank } A = n \Rightarrow \dim(\text{range}(A)) = n$$

-We know $\text{range}(A) \subseteq \mathbb{R}^n$ and $\dim(\mathbb{R}^n) = n$.

$\therefore \text{range}(A) = \mathbb{R}^n$.

GAUSSIAN ELIMINATION EXAMPLE

$$\begin{bmatrix} 2 & 6 & 3 \\ 1 & -2 & \frac{1}{2} \\ -1 & 1 & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 2/5 \end{bmatrix}$$

Row reductions (step-by-step) preserve equalities:

$$\begin{aligned} &\Rightarrow \begin{bmatrix} 2 & 6 & 3 \\ 0 & -1 & 0 \\ -1 & 1 & -1/2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 4 \\ 7/5 \\ 2/5 \end{bmatrix} \\ &\Rightarrow \begin{bmatrix} 2 & 0 & 3 \\ 0 & -1 & 0 \\ -1 & 1 & -1/2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 31/5 \\ 7/5 \\ 2/5 \end{bmatrix} \\ &\Rightarrow \begin{bmatrix} 1 & 0 & 3/2 \\ 0 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 31/5 \\ -7/5 \\ 33/5 \end{bmatrix} \\ &\Rightarrow \begin{bmatrix} 1 & 0 & 3/2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 31/5 \\ -7/5 \\ 8 \end{bmatrix} \\ &\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -29/5 \\ -7/5 \\ 8 \end{bmatrix} \end{aligned}$$

NORM

Given a vector space E , a norm on E is a function $p : E \rightarrow \mathbb{R}^+ = [0, +\infty)$ that satisfies $\forall x, y \in E, \lambda \in \mathbb{R}$:

- $p(x + y) \leq p(x) + p(y)$ (Triangle Inequality)
- $p(\lambda x) = |\lambda| p(x)$ (Homogeneity)
- $p(x) = 0 \iff x = 0$ (The norm is non-degenerate)

p-norm:

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$$

∞ -norm:

$$\|x\|_\infty = \max_{i=1, \dots, n} |x_i|$$

Unit Ball for ∞ -norm:

$$\{x \in \mathbb{R}^2 \mid \max(|x_1|, |x_2|) \leq 1\}$$

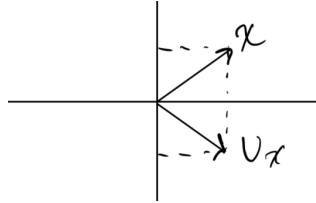
Note: An orthonormal basis of $\mathbb{R}^n$ is an orthogonal basis made of vectors with L2-Norm = 1.

ORTHOGONAL MATRICES

$$UU^T = U^T U = I \quad : \text{ Preserve the L2-Norm of the vector.}$$

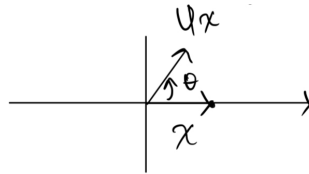
→ In $\mathbb{R}^2$, orthogonal matrices represent Rotations & Reflections.
 → The associated Linear Transforms are called **ISOMETRIES**.

Ex: Reflection w.r.t. the horizontal axis:



$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \\ -x_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Ex: Rotation by angle θ :



$$\begin{bmatrix} x_1 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \cos \theta \\ x_1 \sin \theta \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_1 \\ 0 \end{bmatrix}$$

$\Rightarrow$ A M^x is orthonormal $\iff$ its columns are orthonormal vectors.

CHANGE OF BASIS: DIAGONALIZATION OF MATRIX

Expressing a transformation written in one coordinate system using a different coordinate system.

If we have an $m \times m$ matrix A with m linearly independent eigenvectors $v_1, v_2, \dots, v_m \in \mathbb{R}^m$ with $\lambda_1, \dots, \lambda_m$ as eigenvalues:

$$Av_i = \lambda_i v_i, \quad i = 1, 2, \dots, m$$

Defining $P = [v_1 \ v_2 \ \dots \ v_m]$, $D = \text{diag}(\lambda_1, \dots, \lambda_m)$:

$$\boxed{AP = PD}, \quad A = PDP^{-1}$$

$$AP = [AV_1 \ AV_2 \ AV_3 \ \dots \ AV_m] = [\lambda_1 V_1 \ \lambda_1 V_2 \ \lambda_1 V_3 \ \dots \ \lambda_1 V_m]$$

$$= [V_1 \ V_2 \ V_3 \ \dots \ V_m] \text{diag}(\lambda_1, \dots, \lambda_m)$$

Since the columns of V are linearly independent, V is invertible.

$$A = PDP^{-1}, \quad D = P^{-1}AP \quad \rightarrow \quad \text{Made of Eigenvalues of } A$$

$$\boxed{A^k = PD^k P^{-1}}$$

• Therefore, a square matrix A is said to be diagonalizable over $\mathbb{R}$ when it is similar to a diagonal matrix with coefficients in $\mathbb{R}$, i.e., when there exists an invertible matrix P with coefficients in $\mathbb{R}$ and a diagonal matrix D with coefficients in $\mathbb{R}$

such that:

$$\boxed{A = PDP^{-1}}$$

Theorem: A is diagonalizable over $\mathbb{R} \iff$ there exists a basis of $\mathbb{R}^m$ made of eigenvectors of A .

SPECTRAL THEOREM

If A is a (Real) symmetric $m \times m$ M^x , then $\exists$ an orthogonal $M^x P$ & diagonal $M^x D$ such that:

$$A = PDP^T : D \text{ has the eigenvalues of } A \text{ as its diagonal entries.}$$

– Moreover, $\exists$ an orthogonal basis of $\mathbb{R}^m$ made of eigenvectors of A & all eigenvalues $\in \mathbb{R}$.

POSITIVE DEFINITE MATRICES

- An $m \times m$ $M^x A$ is **P.D.** if for any non-zero $x \in \mathbb{R}^m$:

$$x^T Ax > 0$$

- An $m \times m$ $M^x A$ is **P.S.D.** if:

$$x^T Ax \geq 0$$

Ex: Given an $m \times m$ $M^x A$, the $M^x A^T A$ (Gram Matrix of A) is symmetric & PSD.

$\rightarrow A^T A$ is symmetric:

$$(A^T A)^T = A^T A$$

$\rightarrow A^T A$ is PSD for any $x \in \mathbb{R}^m$:

$$x^T (A^T A)x = (Ax)^T (Ax) = \|Ax\|_2^2 \geq 0$$

POLAR DECOMPOSITION

Given an $m \times m$ square matrix A , we consider $A^T A$ and apply the spectral theorem to say that there exists an orthogonal $M^x V$ and a diagonal $M^x D$ with the eigenvalues of $A^T A$ on its diagonal such that:

$$\boxed{A^T A = VDV^T}, \quad D = \begin{bmatrix} d_1 & & & \\ & d_2 & & \\ & & \ddots & \\ & & & d_m \end{bmatrix}$$

Since $A^T A$ is symmetric PSD, its eigenvalues are ≥ 0 , so:

$$\sqrt{D} = \begin{bmatrix} \sqrt{d_1} & & & \\ & \sqrt{d_2} & & \\ & & \ddots & \\ & & & \sqrt{d_m} \end{bmatrix}, \quad \boxed{\sqrt{A^T A} = V\sqrt{D}V^T}$$

$$\begin{aligned} (\sqrt{A^T A}x)^T (\sqrt{A^T A}x) &= (V\sqrt{D}V^T x)^T (V\sqrt{D}V^T x) \\ &= x^T V\sqrt{D}V^T V\sqrt{D}V^T x = x^T VDV^T x = x^T A^T A x = (Ax)^T (Ax) = \|Ax\|_2^2 \end{aligned}$$

THEOREM: POLAR DECOMPOSITION

For any $m \times m$ $M^x A$, $\exists$ an orthogonal $M^x R$ such that:

$$A = R\sqrt{A^T A}$$

$$A = R\Sigma V^T, \quad \Sigma = \text{diag}(\sigma_1, \dots, \sigma_m), \quad \sigma_i = \sqrt{\lambda_i(A^T A)}$$

→ Since R, V are orthogonal matrices, their product is also orthogonal.

$$R^T R = I, \quad V^T V = I \quad \Rightarrow \quad A = U \Sigma V^T$$

SINGULAR VALUE DECOMPOSITION (SVD)

A linear transformation consists of 3 independent steps: Rotation, Stretching, Rotation.
Therefore, any matrix can be decomposed:

$$A = U \Sigma V^T$$

- Σ contains singular values (scaling factors of the transformation).
- $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n$
- Ignoring smaller singular values gives a good approximation and reduces rank.

For rectangular matrices:

$$A^{m \times n} = U^{m \times m} \Sigma^{m \times n} V^{T \, n \times n}$$

If $m \geq n$:

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_n \\ 0 & 0 & \dots & 0 \end{bmatrix}$$

If $m \leq n$:

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \dots & \sigma_n & 0 & \dots & 0 \end{bmatrix}$$

$$\begin{aligned} \begin{pmatrix} - & - & - & - \\ - & - & - & - \\ - & - & - & - \\ - & - & - & - \end{pmatrix} &= \begin{bmatrix} | & | & | & | \\ U_1 & U_2 & U_3 & U_4 \\ | & | & | & | \end{bmatrix} \begin{pmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \sigma_3 & \\ & & & \sigma_4 \end{pmatrix} \begin{bmatrix} \hline V_1 \\ \hline V_2 \\ \hline V_3 \\ \hline V_4 \end{bmatrix} = \sum_{j=1}^n \sigma_j U_j V_j^T \\ A (\text{rank } 4) &= \sigma_1 \begin{bmatrix} | \\ U_1 \\ | \end{bmatrix} \begin{bmatrix} \hline V_1 \\ \hline \end{bmatrix} + \sigma_2 \begin{bmatrix} | \\ U_2 \\ | \end{bmatrix} \begin{bmatrix} \hline V_2 \\ \hline \end{bmatrix} + \cancel{\sigma_3} \begin{bmatrix} | \\ U_3 \\ | \end{bmatrix} \begin{bmatrix} \hline V_3 \\ \hline \end{bmatrix} + \cancel{\sigma_4} \begin{bmatrix} | \\ U_4 \\ | \end{bmatrix} \begin{bmatrix} \hline V_4 \\ \hline \end{bmatrix} \\ \text{Largest} &\swarrow \quad \nwarrow \text{2nd Largest} \\ \text{RANK 1} &\quad \text{RANK 1} \quad \text{RANK 1} \quad \text{RANK 1} \end{aligned}$$

COMPUTING U , V and Σ

1. Compute $A^T A$, find eigenvectors/eigenvalues:
 V = normalized eigenvectors, $\Sigma = \sqrt{\lambda_i}$
2. Compute AA^T , find eigenvectors/eigenvalues:
 U = normalized eigenvectors.

Thus:

$$A^T A = V \Sigma^T \Sigma V^T, \quad A A^T = U \Sigma \Sigma^T U^T$$

$$A = U \Sigma V^T$$

Low Rank Approximation Theorem

Consider $1 \leq R \leq r \leq n$
→ r : # of singular values
→ n : # of non-0 singular values

Denote

$$A_R = \sum_{j=1}^R \sigma_j U_j V_j^T$$

then

- $\|A - A_R\|_2 = \sigma_{R+1}$
- If B is an $m \times m$ M^r with $\text{rank}(B) = R$, then

$$\|A - B\|_2 \geq \|A - A_R\|_2 \geq \sigma_{R+1}$$

Trace of a Matrix

For a sq. Mx, the trace is the sum of the diagonal elements

$$\text{Tr}(A) = \sum_{i=1}^n a_{ii}$$

1. $\text{Tr}(A + B) = \text{Tr}(A) + \text{Tr}(B)$ (Linear Transformation)
2. $\text{Tr}(cA) = c \cdot \text{Tr}(A)$
3. $\text{Tr}(AB) = \text{Tr}(BA)$

Theorem: $\text{Tr}(A) = \sum_{i=1}^n \lambda_i$, where λ_i are the eigenvalues of A .

Matrix Norms

→ Given a vector norm $\|\cdot\|$, we define norm of an $m \times m$ matrix A as:

$$\|A\| = \sup_{x \in \mathbb{R}^m, x \neq 0} \frac{\|Ax\|}{\|x\|}$$

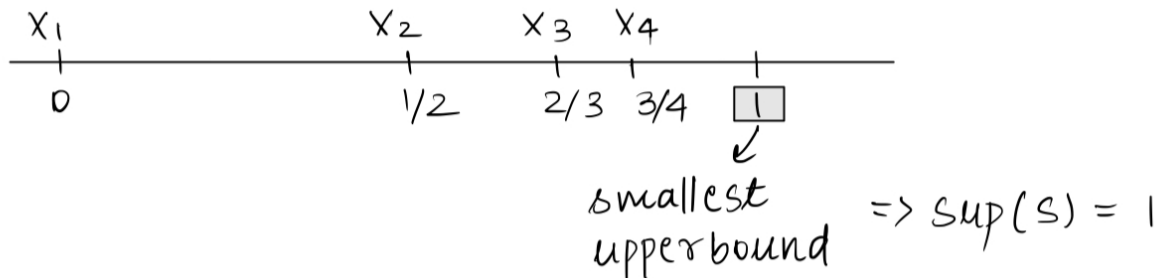
$$S = \left\{ \frac{\|Ax\|}{\|x\|}, \forall x \in \mathbb{R}^m, x \neq 0 \right\}$$

Parenthesis about the Supremum of a Set S

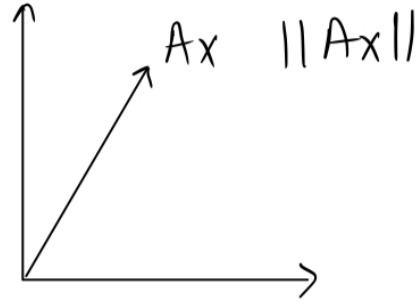
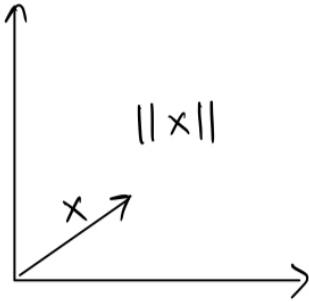
The supremum of S is the smallest of the upper bounds of the set S . If the set has a maximum, the sup coincides with the maximum.

Example:

$$S = \{x_m \in \mathbb{R}, \text{ s.t. } x_m = 1 - 1/m, m \in \mathbb{N} \setminus \{0\}\}$$



Geometrical Interpretation in $\mathbb{R}^2$



$$\frac{\|Ax\|}{\|x\|} : \text{Expansion Factor} \Rightarrow \|A\| : \text{Largest Expansion Factor}$$

Property:

$$\|A\| = \max_{\|x\|=1} \|Ax\|$$

Proof:

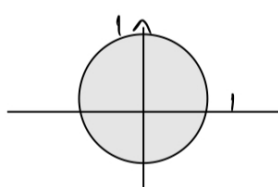
$$\|A\| = \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \sup_{x \neq 0} \frac{1}{\|x\|} \|Ax\| = \sup_{x \neq 0} \left\| \frac{Ax}{\|x\|} \right\|$$

$$\text{Let } y = \frac{x}{\|x\|}, \text{ then } \|y\| = 1 \rightarrow \|A\| = \sup_{\|y\|=1} \|Ay\|$$

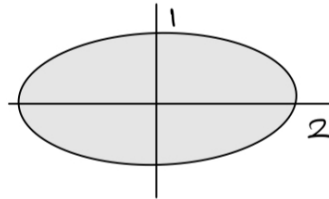
Example:

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \text{ Calculate } \|A\|_2$$

$$\|A\|_2 = \sup_{\|x\|_2=1} \|Ax\|_2 = \max_{\|x\|_2=1} \|Ax\|_2$$



$$\{x \in \mathbb{R}^2, \|x\|_2 = 1\}$$



$$\{Ax, x \in \mathbb{R}^2, \|x\|_2 = 1\}$$

$$\Rightarrow \|A\|_2 = 2$$

Formulas to Compute L^1 and L^2 Norms of a Matrix

- $\|A\|_2 = \sup_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2} = \sqrt{\text{spectral radius}(A^T A)}$

where the **Spectral Radius** of a matrix is the maximum of the absolute values of the eigenvalues of the matrix.

Example:

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad A^T A = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\Rightarrow \text{spectral radius}(A^T A) = 4 \Rightarrow \|A\|_2 = 2$$

Note: If A is symmetric $\Rightarrow \|A\|_2 = \text{spectral radius}(A)$

- $\|A\|_\infty = \sup_{x \neq 0} \frac{\|Ax\|_\infty}{\|x\|_\infty} = \max_{1 \leq i \leq m} \sum_{j=1}^n |a_{ij}|$
(Sum of the absolute values of the entries of the i -th row)

Example:

$$A = \begin{pmatrix} -1 & 2 & 3 \\ -3 & 1 & 4 \\ 4 & 1 & 2 \end{pmatrix}$$

$$\Rightarrow \|A\|_\infty = \max\{6, 8, 7\} = 8$$

- $\|A\|_1 = \sup_{x \neq 0} \frac{\|Ax\|_1}{\|x\|_1} = \max_{1 \leq j \leq m} \sum_{i=1}^m |a_{ij}|$
(Sum of the absolute values of the entries of the j -th column)

Example:

$$A = \begin{pmatrix} -1 & 2 & 3 \\ -3 & 1 & 4 \\ 4 & 1 & 2 \end{pmatrix}$$

$$\Rightarrow \|A\|_1 = \max\{8, 4, 9\} = 9$$

Properties

- $\|Ax\| \leq \|A\| \|x\|$ for any matrix A & vector x
- $\|AB\| \leq \|A\| \|B\|$ for any two matrices A, B

$$\rightarrow \|A\| = \sup_{x \neq 0} \frac{\|Ax\|}{\|x\|}$$

Condition Number of a Matrix

→ Consider the linear system $Ax = b$. Suppose there is a perturbation causing an error δb on the data b :

$$b \rightarrow b + \delta b$$

Then the solution x will be perturbed as:

$$x \rightarrow x + \delta x$$

$$A(x + \delta x) = b + \delta b$$

$$x + \delta x = A^{-1}(b + \delta b) = A^{-1}b + A^{-1}\delta b$$

$$\Rightarrow \delta x = A^{-1}\delta b \quad \Rightarrow \quad \|\delta x\| = \|A^{-1}\delta b\| \leq \|A^{-1}\| \|\delta b\|$$

Relative Error:

$$\frac{\|\delta x\|}{\|x\|} \leq \|A^{-1}\| \frac{\|\delta b\|}{\|x\|}$$

Estimating $\frac{1}{\|x\|}$: $b = Ax \Rightarrow \|b\| = \|Ax\| \leq \|A\| \|x\|$

$$\Rightarrow \frac{1}{\|x\|} \leq \frac{\|A\|}{\|b\|}$$

$$\Rightarrow \frac{\|\delta x\|}{\|x\|} \leq \underbrace{\|A^{-1}\| \|A\|}_{\text{Condition Number of } A} \underbrace{\frac{\|\delta b\|}{\|b\|}}_{\text{Relative Error on the input/data}}$$

→ $\boxed{\text{cond}(A) = \|A\| \|A^{-1}\|}$: The worst Expansion Factor of the error.

If $\text{cond}(A)$ is large, even a small error on the input b can give a large error on the solution x . If $\text{cond}(A)$ is large, A is said to be *Ill Conditioned*.

Properties:

- $\frac{1}{\text{cond}(I)} = 1 \Rightarrow \text{cond}(I) = \|I\| \|I^{-1}\| = \|I\| \|I\| = 1$

$$\|I\| = \max_{\|x\|=1, x \in \mathbb{R}^m} \|Ix\| = \max_{\|x\|=1} \|x\| = 1$$

$$\bullet \quad \text{cond}(A) \geq 1 \quad \Rightarrow \quad AA^{-1} = I \quad \Rightarrow \quad \|AA^{-1}\| = \|I\| = 1$$

$$1 = \|AA^{-1}\| \leq \|A\| \|A^{-1}\| = \text{cond}(A)$$

Note: The condition number $\text{cond}_2(A) = \|A\|_2 \|A^{-1}\|_2$ of an orthogonal $M^x P$ is 1.
 $\therefore$ Orthogonal matrices are very well conditioned.

Proof

$$\text{cond}_2(P) = \|P\|_2 \|P^{-1}\|_2 = \|P\|_2 \|P^T\|_2$$

$$\|P\|_2 = \max_{\|x\|_2=1, x \in \mathbb{R}^n} \|Px\|_2 = \max_{\|x\|_2=1} \|x\|_2 : \text{Orthogonal matrices preserve norm (Rotation).}$$

$$\Rightarrow \|P\|_2 = 1$$

$$\rightarrow \text{Similarly, } \|P^T\|_2 = 1 \Rightarrow \boxed{\text{cond}_2(P) = 1 \cdot 1 = 1}$$

Property: Determinant of an orthogonal matrix = ± 1

$$\det(PP^T) = \det(I) = 1 \quad \Rightarrow \quad \det(P) \cdot \det(P^T) = \det(P)^2 = 1$$

$$\Rightarrow \det(P) = \pm 1$$

Relation Between the Determinant and SVD

$$\text{Let } A^{n \times n} = U \Sigma V^T$$

$$\Sigma = \begin{pmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \end{pmatrix}$$

$$|\det(A)| = |\det(U)| |\det(\Sigma)| |\det(V^T)|$$

Since $|\det(U)| = |\det(V^T)| = 1$ for orthogonal U, V ,

$$\Rightarrow \boxed{|\det(A)| = |\sigma_1 \sigma_2 \cdots \sigma_n| = \sigma_1 \sigma_2 \cdots \sigma_n}$$

(Singular values ≥ 0)

Relation Between L^2 -Norm and Condition Number

$$\text{Let } A = U \Sigma V^T$$

$$\|A\|_2 = \|U \Sigma V^T\|_2 = \|\Sigma V^T\|_2$$

Proof: For a general $M^x B$ & orthogonal $M^x U$:

$$\|UB\|_2 = \max_{\|x\|_2=1, x \in \mathbb{R}^n} \|UBx\|_2 = \max_{\|x\|_2=1} \|Bx\|_2 = \|B\|_2$$

$$\Rightarrow \|UB\|_2 = \|B\|_2 \text{ (since } U \text{ is orthogonal: preserves } L^2 \text{ norm of vector).}$$